



Department of Computer Science and Engineering

Academic Year 2022– 2023 (Odd Semester)

Degree, Semester & Branch: I Semester B.Tech Information Technology

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member (s): Dr. R. Venkatesh, AP(SG)/CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV / List, Tuples, Dictionaries
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO11
- **Activity Chosen:** Crossword Puzzle
- **Justification:**
 - A crossword puzzle is a word puzzle that usually consists of a white and black-shaded squares. By solving clues that lead to the answers, the goal is to fill the white squares with letters, forming words or phrases.
 - List, Tuples and Dictionary are the common data structures used in python. It is important to know about the operations and properties of list, tuples and dictionaries
 - So, a crossword puzzle activity is chosen to have a better understanding of the list, tuple and dictionary concept by finding the suitable word according to the question
- **Time** Allotted for the Activity: 15 Minutes
- **Details of the Implementation:**
 - Crossword puzzle of 10 questions were given to each students.
 - The students were understanding the questions, patterns and completed the puzzle as shown in Fig 1
 - After completion of the puzzle activity, the students were discussed among them and finally the faculty member discussed the answer to make the students aware of the correct answer as shown in Fig 2

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	3	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)



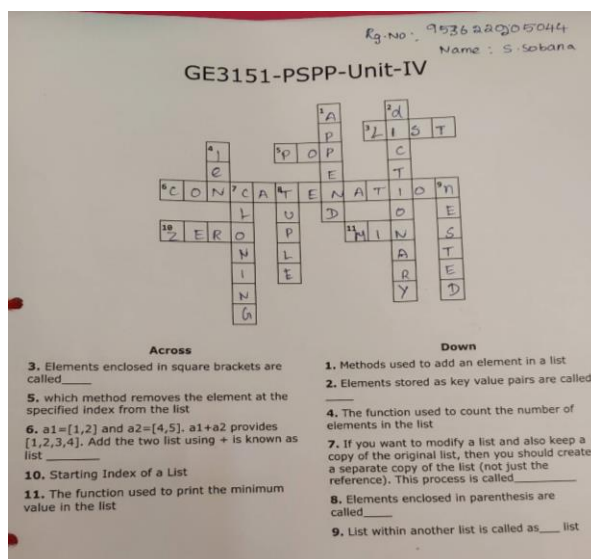
• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	Students will be able to understand the concept of python data structure	Students will be able to choose the appropriate data structure while approaching the problem	Students will be able to provide solution for the problem by applying concept of list, tuple and dictionary	As the student involved in the activity as a team of two, the team work skill improved	Students communication skill will be improved as they are discussing the answers with peers	The problem solving skill earned through this activity helps the students in solving various problems

• Images / Screenshot of the practice:



Fig 1: Students Solving Crossword Puzzle





- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students said that the crossword puzzle activity was very interesting and they were able to identify the appropriate concepts related to list, tuple and dictionary.
- They said the it will be helpful while preparing for examination

- ❖ ***Benefit of the practice:***

- Student's understanding of concepts lists, tuples and dictionaries is improved.
- Vocabulary of the terms related to list, tuple and dictionary is improved
- This activity helps to test the level of understanding of the students

- ❖ ***Challenges faced in implementation:***

- Few students found it difficult to complete the puzzle
- Planned for 15 minutes, but unable to be completed within 15 minutes, it will take an extra 5 minutes to complete.
- Students just might not have the necessary knowledge to complete crossword puzzles

References:

- <https://www.ritrjpm.ac.in/images/computer-science/Crosswords-MCAP-GM.pdf>
- <http://www.classtools.net/crossword/>
- <https://www.activityvillage.co.uk/crosswords>

Signature of Faculty Member

HOD-CSE